

Mihir Garimella

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EDUCATION

Columbia University, B.S. in Computer Science & Applied Mathematics

Expected May 2029

- C. P. Davis Engineering Scholar, Nonlinear Controls Lab, Columbia Robotics Club (Lead), Clefhangers A Cappella
- Coursework: Data Structures, Computer Vision I, Computer Systems, Linear Algebra, Advanced Programming

EXPERIENCE

NASA Jet Propulsion Laboratory/Caltech, Skyfall Mars Helicopter Team

Jun 2026 - Aug 2026

Guidance, Navigation, & Controls Intern

Pasadena, CA

- Reduced terrain-relative localization error by 20% by matching descent imagery against onboard terrain maps within dynamically constrained search areas, achieving 20m accuracy at 10 Hz via normalized cross-correlation template matching
- Built and flight-tested microsecond-precise helicopter deployment platform pose characterization system (RasPi 5, IMU, GPS) to provide continuous, accurate vehicle pose ground truth under extreme Mars entry-descent aerodynamic regimes

Rotor Technologies

Jun 2024 - Aug 2025

Perception & Software Engineering Intern

Nashua, NH

- Improved short-range landing accuracy by 75% by developing a vision-based helicopter precision landing system, progressing from lightweight CNN-based trailer-bed detection to ArUco marker detection and PnP pose estimation, deploying it in a flight-ready build
- Cut computer provisioning time by 40% by authoring shell scripts to automate Ubuntu NUC provisioning, simulation deployment, time synchronization, and environment validation

Dartmouth Reality & Robotics Lab

Jun 2023 - Jun 2024

Research Assistant

Hanover, NH

- Developed obstacle avoidance system for a low-cost water surface vehicle running ArduPilot waypoint navigation, using YOLOv8s on a Jetson Nano for monocular obstacle detection and a reactive controller to evade obstructions
- Co-authored autonomous surface vehicle research, evaluating multi-object collision avoidance and motion planning using LiDAR, stereo-vision, and IMU through aquatic trials with dynamic obstacles for water-quality monitoring

PROJECTS

UAV-UGV Coordination for Object Discovery, Localization & Retrieval (Nonlinear Controls Lab)

- Implemented a stereo-vision pipeline for drone surveying, using HSV color segmentation, connected-component analysis, and disparity-based depth estimation to detect and localize objects in the UAV camera frame during simulated flight
- Built an aerial-to-ground coordination system to transform UAV detections into global target coordinates and autonomously route the UGV (simulated Boston Dynamics Spot) to target locations via linear waypoint interpolation

Modeling of 3D Pre-Operative Brain Tumor Volume in Glioblastoma Patients

- Built a multi-model (U-Net, ConvLSTM) computer vision pipeline to segment and forecast brain tumor growth in patient MRI scans 6 months in advance, achieving 0.86 Dice segmentation accuracy in under 2 minutes of inference
- Developed GPU-optimized PyTorch workflow with a custom loss function, reducing U-Net training time from 30hrs to 10hrs on an NVIDIA RTX 3070
- Selected Presentation at Stanford Neuroscience Forum

Students x Artificial Intelligence Outreach Organization (HSxAI)

- Founded HSxAI, a public video series connecting 50k+ students to researchers working on cutting-edge AI and robotics
- Organized 30 events with figures in AI, invited to cover research presented at IROS as a student journalist

SKILLS

Languages & Frameworks: Python, C++, Java, TypeScript, Bash, SQL, CUDA, PyTorch, TensorFlow, Keras, HuggingFace

Tools & Systems: Git, Linux, ROS, React, Docker, CI/CD, LiDAR, Camera/Vision Systems, IMU, GPS

Math & Statistics: Linear Algebra, Probability, State Estimation, Machine Learning, Deep Learning